# Irrigation strategies under contrasting water-storage and rainfall conditions in the North China Plain

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## Abstract

Efficient irrigation allocation is essential for sustaining grain production while limiting water consumption in the North China Plain wheat–maize rotation. How spatial targeting redistributes production losses, and whether regional water-storage information improves rainfall-based irrigation decisions, remain unresolved. Continuous process-based crop simulations compared uniform and targeted allocations under equal irrigation budgets and evaluated annual storage–rainfall strategies against matched rainfall-only controls during 2014–2025. Growth parameters were fitted to multisite observations; water-use parameters were calibrated against Wuqiao treatments in 2016–2018 and retrospectively tested in 2019. At a 50% irrigation reduction, targeting retained 0.73 t $ha^{-1}$ $yr^{-1}$ more simulated combined dry grain than uniform cuts across 9.87 million rotation hectares, equivalent to 7.22 million t $yr^{-1}$ or 4.87% of uniform production. Mean production was lower than under uniform cuts on 37.60% of mapped area, while crop-plus-fallow evapotranspiration increased by 20.76 mm $yr^{-1}$ (2.05 $km^3$ $yr^{-1}$). Across nine years with eligible Gravity Recovery and Climate Experiment (GRACE) observations, strategies selected at 95% and 98% grain targets retained 96.27% and 99.63% of conventional production, respectively. The 95% strategy reduced irrigation by 130.72 mm and evapotranspiration by 36.84 mm per rotation. Matched storage–rainfall and rainfall-only strategies selected identical quotas and produced identical outcomes. These conditional results support joint production and consumptive-use constraints and local production safeguards, while showing that additional storage information provides no decision benefit within the tested quota design.

## Keywords

winter wheat–summer maize rotation; irrigation management; crop modeling; GRACE; water availability; consumptive water use

## 1. Introduction

Irrigation management in the North China Plain must reconcile grain production with water-resource constraints across the winter wheat–summer maize rotation (Yang et al., 2022). Wheat grows through a relatively dry season, whereas maize receives more summer rainfall; soil-water reserves connect their responses to irrigation (Yang et al., 2024). Long-term well-watered observations show that rising grain yield has accompanied increasing actual evapotranspiration (ET), despite comparatively stable reference ET (Zhang et al., 2011). Maintaining production with less irrigation therefore requires attention to both the water delivered to fields and the water consumed over the complete rotation.

The timing and location of irrigation supply determine its marginal production return. A sixteen-year experiment in 2003–2018 compared seven wheat–maize irrigation schedules and found diminishing production gains as irrigation increased (Yang et al., 2022). A four-year Wuqiao experiment also showed that precipitation conditions influenced the preferred wheat schedule and grain production in the subsequent maize crop (Yang et al., 2024). These responses reflect the sequence of crop demand, rainfall and soil-water carry-over across both seasons (Yang et al., 2024; Yang et al., 2022). At regional scale, irrigation requirements vary among locations and hydrological years (Ren et al., 2022; Wu et al., 2019). Crop-model optimization and irrigation-allocation studies provide approaches for using this heterogeneity to guide management (Jiang et al., 2016; Xiao et al., 2024). Under a common water budget, targeting locations with larger marginal grain returns can improve regional production retention, while changing the distribution of production losses.

Assessing these allocation consequences requires consistent crop and water accounting. Field delivery, ET, return flows and storage changes describe different quantities, so a reduction in irrigation does not directly establish an equivalent reduction in consumption (Grafton et al., 2018; Perry et al., 2009). Process-based applications of APSIM and RZ-SHAW in the North China Plain connect crop production with water productivity, field water balance and ET (Chen et al., 2010; Fang et al., 2014). However, transpiration formulation remains a major source of variation in simulated wheat water use (Cammarano et al., 2016). Phenology, canopy, biomass, grain and ET observations

therefore constrain complementary processes, while documented irrigation treatments assess their combined response to water supply. Continuous simulation retains the soil-water carry-over needed to compare complete rotation histories.

Spatial allocation determines where irrigation is supplied, while antecedent hydrological conditions can inform the annual irrigation budget. Recent rainfall describes water input at the crop-system scale, whereas GRACE total water storage integrates several terrestrial stores over a regional footprint (Save et al., 2016; Scanlon et al., 2016). Recent aquifer recovery under water transfers, abstraction management and changing climate illustrates the broader context of regional storage variation (Long et al., 2025; Xu et al., 2026). Storage information may thus characterize conditions beyond local rainfall. Its practical value for irrigation nevertheless depends on whether it changes the selected action or improves the resulting grain–water outcome relative to rainfall-based decisions.

Two connected management questions remain unresolved. First, it is unclear how much grain spatial targeting can retain under an identical irrigation budget, and what consumptive-water costs and persistent local production losses accompany that regional benefit. Rotation-wide accounting must retain soil-water carry-over to assess these outcomes together. Second, the additional decision value of regional storage information beyond antecedent rainfall remains uncertain. A storage indicator can distinguish hydrological conditions without changing an irrigation choice. Comparisons using the same production targets, candidate quotas and monitoring availability are needed to isolate its contribution to management outcomes.

The objectives were to (i) quantify rotation grain production, crop-plus-fallow ET and profile drainage under uniform and spatially targeted irrigation allocations with common regional water budgets; (ii) characterize the spatial distribution, magnitude and annual persistence of allocation-related grain gains and losses; and (iii) assess whether antecedent regional storage information changes annual irrigation quotas and grain–water outcomes relative to rainfall-only controls under matched selection and monitoring conditions. Continuous Open Crop Model simulations of the North China Plain wheat–maize rotation, parameterized using multisite crop observations and documented irrigation treatments, provide the basis for these conditional comparisons during 2014–2025.

# 2. Materials and methods

## 2.1. Overall workflow

The workflow links crop-model calibration and evaluation to regional irrigation-management comparisons through four steps (Figure 1). First, crop development and growth are calibrated and evaluated, and crop water-use parameters are fitted to irrigation-treatment responses and tested retrospectively. The resulting parameters remain fixed for the management comparisons. Second, continuous winter wheat–summer maize simulations quantify crop responses to five irrigation levels, with soil water carried through successive crops and intervening fallows.

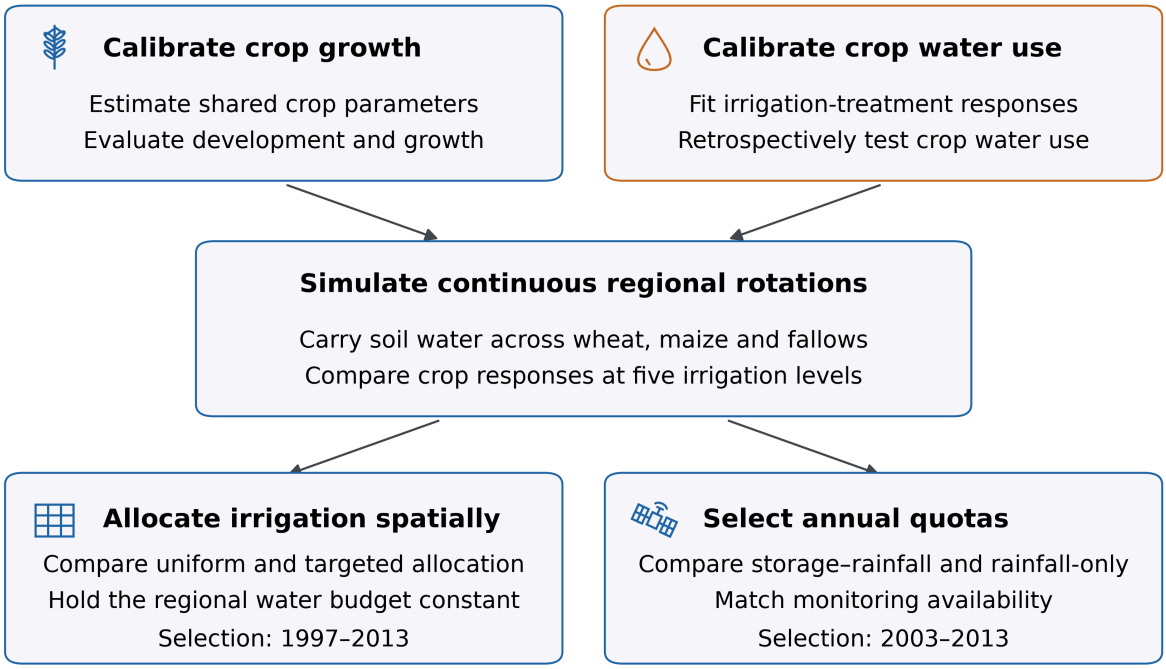



**Figure 1. Study workflow. Crop growth and water-use calibration and evaluation precede continuous regional rotation simulations. Historical crop responses support spatial irrigation allocation under equal water budgets and annual quota selection from antecedent water availability. The strategies are compared during 2014–2025 using rotation grain production, field irrigation, crop-plus-fallow ET, profile drainage and the spatial distribution of production gains and losses.**

Third, historical simulated responses support two complementary irrigation decisions: where to allocate a regional water budget and how much irrigation to supply in a given year. Spatial targeting and uniform reductions share identical regional irrigation budgets. Annual quota selection compares antecedent storage–rainfall classes with rainfall-only controls under matched production targets, candidate quotas and monitoring availability. Finally, the

selected strategies are compared during 2014–2025 using rotation grain production, field irrigation, crop-plus-fallow evapotranspiration (ET), profile drainage and the spatial distribution and persistence of production gains and losses.

## 2.2. Study area

The study area encompasses the winter-wheat–summer-maize lowlands of the North China Plain within 112–121.5°E and 32–41°N, extending from Beijing, Tianjin and Hebei through Shandong and Henan to the northern Anhui and Jiangsu plains (Figure 2) (Wu et al., 2019). The landscape consists of extensive alluvial lowlands, with piedmont plains along the Taihang and Yanshan mountains in the west and north and lower coastal plains toward the east (Long et al., 2025; Xu et al., 2026). Its low relief supports intensive agricultural production, while climatic and soil-water differences contribute to spatial variation in irrigation requirements (Ren et al., 2022).

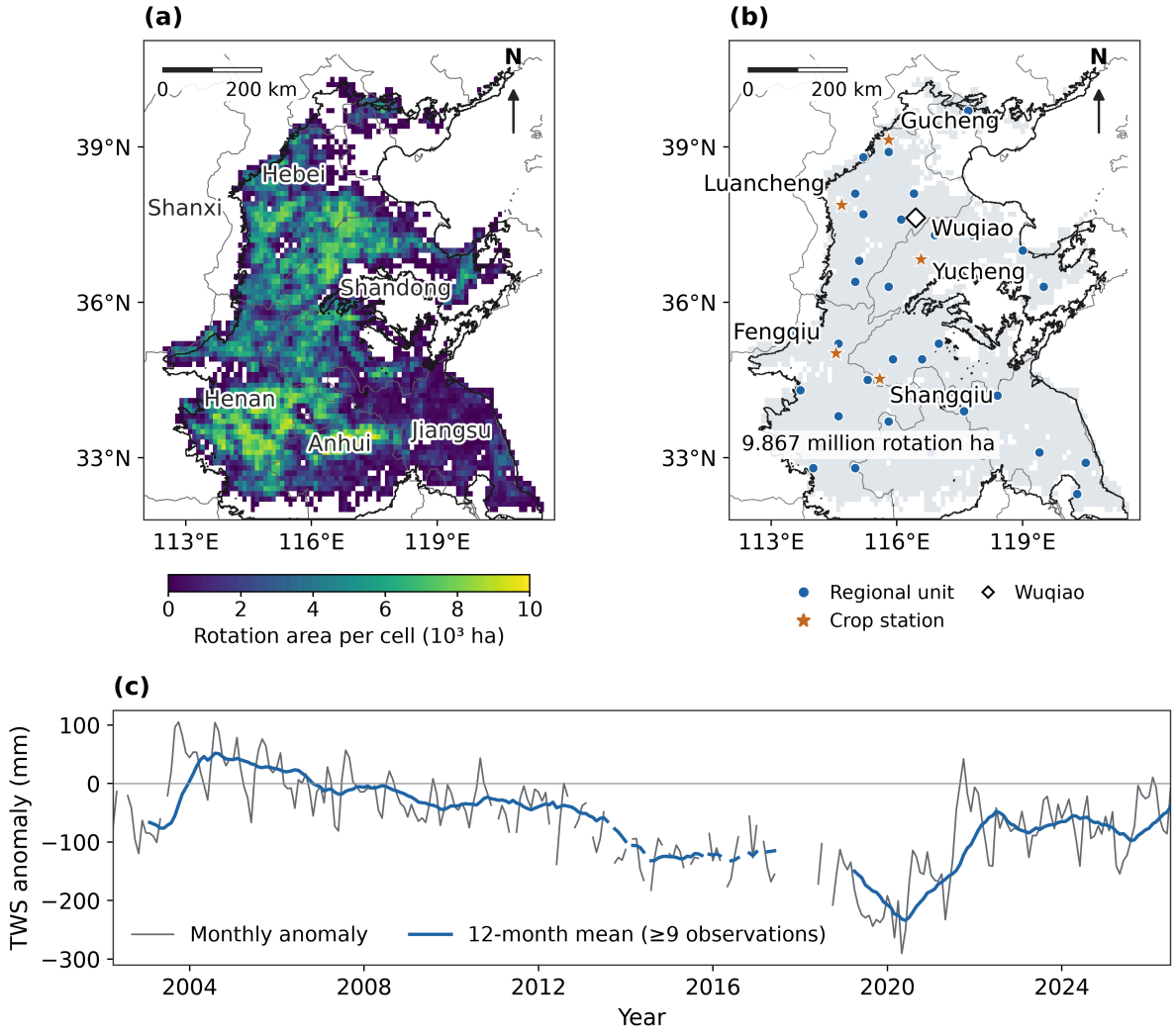



**Figure 2. Study domain, rotation area, observation sites and regional water storage. (a) Mapped 2020 wheat–maize rotation area. (b) Five crop-observation sites, the Wuqiao irrigation experiment and 32 regional simulation units. (c) Observed CSR monthly terrestrial-water-storage anomalies and a 12-calendar-month mean requiring at least nine observations. Missing months remain gaps. Thin grey lines show provincial boundaries; black outlines delimit the operational physiographic study domain. Maps use an equirectangular projection (36°N, 117°E). Provincial boundaries: Natural Earth Admin 1 v5.1.1 (Natural Earth, n.d.). North arrows indicate geographic north; scale bars use WGS84 ellipsoidal ground distances at their displayed latitude.**

The region has a continental monsoon climate, with cold, dry winters and warm, rainy summers; regional assessments report mean annual temperatures of approximately 10–15°C (Ren et al., 2022). Rainfall is concentrated in the summer monsoon, whereas spring is frequently dry and precipitation varies substantially among years and locations (Wu et al., 2019; Xu et al., 2026). Winter wheat is sown in autumn and harvested in early summer, followed by summer maize harvested in autumn. Wheat therefore grows largely outside the rainy season and commonly requires supplementary irrigation before sowing and during spring growth. Maize overlaps the rainy

season, with irrigation demand depending more strongly on rainfall distribution and dry-year conditions. The two crops share a soil-water reserve, so irrigation in wheat can affect the water available to subsequent maize (Yang et al., 2024; Yang et al., 2022).

Groundwater has historically supplied much of the irrigation deficit, particularly in the northern plain, where intensive pumping for agriculture and urban demand caused prolonged groundwater decline (Xu et al., 2026). The aquifer system includes both unconfined and deeper confined units, with substantial spatial differences in groundwater depth and depletion (Long et al., 2025). Irrigation water availability also reflects access to river diversions and the allocation of transferred surface water. Since 2020, groundwater recovery has been documented across much of the northern North China Plain following the expansion of South-to-North Water Diversion supplies, reduced abstraction and water-saving measures, with wet years reinforcing recovery (Long et al., 2025; Xu et al., 2026). This transition places irrigation management within a changing water-supply setting: maintaining recovery requires attention to consumptive use and seasonal demand as well as the volume delivered to fields.

## 2.3. Field observations and environmental data

### 2.3.1. Multisite crop observations

Winter-wheat and summer-maize observations supported crop-growth parameter estimation and comparisons across five North China Plain sites. Long-term monitoring at Fengqiu, Luancheng, Shangqiu and Yucheng (Fengqiu Station, 2020a; Luancheng Station, 2020; Shangqiu Station, 2021; Yucheng Station, 2025a; Yucheng Station, 2021) was complemented by the Gucheng maize sowing-date experiment (Geng et al., 2023). The National Ecosystem Science Data Center archives contain intermittent crop seasons during 1998–2022; exact site–crop–year coverage and dataset identifiers are given in Tables S1 and S2.

Whole site-years retained their calibration or retrospective-evaluation assignments (Figure 3). The archive comprised 132 crop–plot–season trajectories, of which 127 contained eligible LAI, biomass, ET or harvest observations (Table 1). Daily ET and antecedent soil moisture came from the Fengqiu soil-water archive (Fengqiu Station, 2020b); ET was also available from the Yucheng water-environment archive (Yucheng Station, 2025b). Grain and harvest-biomass samples were confined to Yucheng. Within-season measurements and quadrats describe temporal dynamics and sampling variation rather than independent growing seasons.

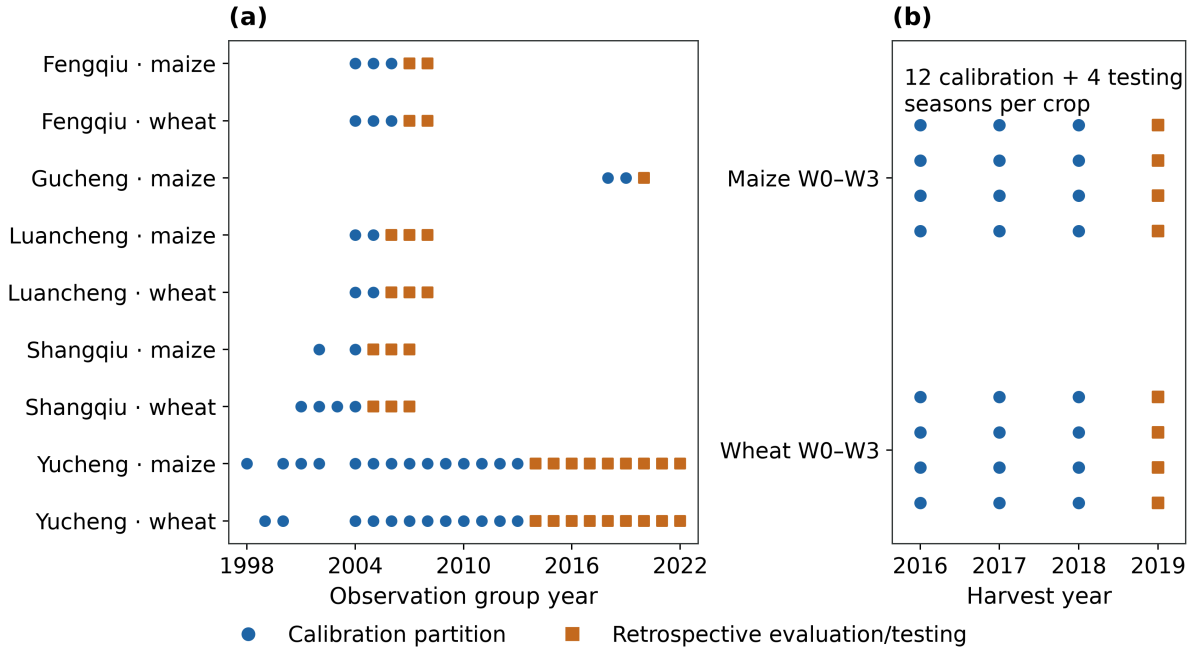



**Figure 3. Evidence partitions. (a) Whole station site-years retain their original calibration and retrospective-evaluation partitions; dates denote observation-group years. All five eligible sites are included. (b) Four Wuqiao treatment means per crop-year provide 12 calibration seasons in 2016–2018 and four retrospective testing seasons in 2019. Station comparisons retain earlier growth-fitting history and unconfirmed complete irrigation management.**

Observations were matched to predictions by crop, plot, quantity and sampling date or averaging window. Screening excluded wheat LAI above 7 $m^2\ m^{-2}$ and duplicated maize harvest groups of unresolved authenticity; raw values and exclusion reasons were retained. Complete plot-specific irrigation histories remain unconfirmed, so station comparisons are conditional on archived management assumptions and have no weight in the water-response fit based on documented Wuqiao treatments (Section 2.3.2). Measurement definitions, screening details and antecedent soil-moisture initialization are specified in Supplementary Section S1.

**Table 1. Eligible station-observation records by site and variable.**

| Site | LAI | Biomass | Daily ET | Yield | Harvest biomass |
|---|---|---|---|---|---|
| Fengqiu | 159 | 163 | 962 | 0 | 0 |
| Gucheng | 216 | 0 | 0 | 0 | 0 |
| Luancheng | 123 | 165 | 0 | 0 | 0 |
| Shangqiu | 114 | 148 | 0 | 0 | 0 |
| Yucheng | 183 | 67 | 4713 | 192 | 192 |

Counts denote individual observations, not independent experiments. Station comparisons retain inherited growth-fitting history and unconfirmed complete management. Yucheng grain and harvest biomass contain six dry-matter samples per crop-year; the duplicated 2012 and 2017 maize groups remain excluded.

### 2.3.2. Field irrigation experiment

The Wuqiao winter-wheat–summer-maize experiment (37.630°N, 116.444°E) comprised four wheat irrigation regimes during 2015–2019 (Yang et al., 2024). W0 received no wheat irrigation; W1 received 75 mm before sowing; W2 received an additional 75 mm at jointing; and W3 received a further 75 mm at anthesis. Total wheat irrigation was therefore 0, 75, 150 and 225 mm, respectively. Treatments had three field replicates and a common nitrogen input of 158.4 kg $ha^{-1}$ per crop.

Maize irrigation was identical across treatments within each year: no irrigation in 2016 and 2018, and 75 mm on 18 June in 2017 and 2019. Maize W0–W3 labels consequently denote preceding wheat irrigation rather than different maize irrigation schedules. The treatment structure distinguishes direct wheat irrigation contrasts from maize responses associated with preceding wheat management.

Crop–treatment–year observations comprised seasonal ET, grain yield and dry aboveground biomass. ET was estimated from rainfall, irrigation and depletion of 0–2 m soil-water storage, assuming negligible runoff and drainage; grain yields were standardized to 13% moisture (Yang et al., 2024). Observations from 2016–2018 constrained water-response coefficients, while 2019 remained outside fitting and parameter selection. Prior inspection during model development makes the latter comparison retrospective (Figure 3). Simulation initialization, ET accounting and grain digitization are detailed in Supplementary Section S4.

### 2.3.3. Meteorological, soil, crop-rotation and terrain data

Daily AgERA5 v2.0 weather data on the common 0.1° grid supplied minimum and maximum air temperature, precipitation, solar radiation, reference evapotranspiration, vapor pressure and 10-m wind speed for 1996–2025 (Copernicus Climate Change Service, 2020). Reference evapotranspiration ($ET_0$) entered the water-balance module directly.

Native 250 m SoilGrids v2.0 maps of texture, bulk density, organic carbon and coarse fragments were averaged to 0.01°, then aggregated over mapped rotation pixels within 0.1° meteorological cells, for six depth intervals between 0 and 2 m (ISRIC — World Soil Information, 2020). Hydraulic properties were derived using Saxton–Rawls

equations with bulk-density and gravel adjustments (Saxton and Rawls, 2006). Organic matter was approximated using an assumed organic-matter/organic-carbon ratio of 1.724, and air-dry water content was assumed to equal half the wilting-point value. Hydraulic derivation is detailed in Supplementary Section S2.

The 500 m ChinaCP map for 2020 identified winter-wheat–summer-maize rotation pixels, which were paired with the corresponding CPM cropland-percentage map (Qiu et al., 2022). Physical classified-cell areas were corrected for projection scale and multiplied by cropland percentages to obtain rotation-area weights. CPM values were sampled at ChinaCP pixel centers after coordinate transformation because the source grids had different origins. Pixel centers assigned these weights to the study area and 0.1° meteorological cells. The rotation footprint was 9,866,733 ha before soil screening; three cells with wholly missing soil profiles, totaling 131 ha, were excluded, leaving 9,866,602 ha in 3,641 meteorological cells. The fixed 2020 footprint supplies a common area basis across climate years, so modeled totals describe climatic and management responses rather than historical production trends. The 2015–2021 area series and common-extent comparison with CCD2020 quantify mapping sensitivity in Supplementary Section S3.

SRTMGL1 v003 elevation was bilinearly resampled to 0.001°, and slope was derived at that support (NASA JPL, 2013). Aligned 10 × 10 cell means provided elevation and slope on a 0.01° grid. Catchment screening, elevation ≤200 m and slope ≤2° delineated a connected lowland study area of 404,658 km² within the geographical extent described in Section 2.2. Geographical and terrain-threshold sensitivities are reported in Supplementary Section S3. Crop-area weights and aggregated soil profiles were linked to daily weather through common 0.1° cell identifiers.

### 2.3.4. Terrestrial-water-storage observations

CSR RL06.3 monthly GRACE/GRACE-FO total-water-storage anomalies were aggregated using physical cell areas and the official CSR land mask (Save, 2020; Save et al., 2016). The land-mask overlap covers 395,969 km² within the 404,658-km² study area; excluded coastal supports account for 2.15% of its area. The archive contains 259 observed monthly solutions from April 2002 to July 2026 relative to the 2004–2009 mean. Source values in centimeters were converted to millimeters, and missing months remain missing. A 12-calendar-month moving mean requires at least nine observations. The 0.25° product grid is finer than the effective GRACE observation scale, so one regional storage indicator is shared across all simulation units (Save, 2020; Save et al., 2016). Soil, surface and other stores were not subtracted; the indicator represents total water storage rather than groundwater alone.

The corresponding aggregation without the CSR land mask is retained as a sensitivity analysis. Regional storage changes provide hydrological context for water-management pressures, with no causal attribution to particular irrigation treatments, water transfers or groundwater regulations (Long et al., 2025; Long et al., 2020).

## 2.4. Crop model description

Open Crop Model (OCM) (Zhao et al., submitted) couples crop growth and a layered soil-water balance at a daily time step (Figure 4). Wheat development combines temperature, vernalization and photoperiod responses; maize development follows thermal time. Development stage (DVS) is anchored at emergence, flowering and maturity. Germination begins when water above the wilting-point storage is available in the represented layer containing the seed; subsequent thermal development starts from that date. Stage coefficients remain fixed, while the soil-water state can alter establishment timing. Recorded or prescribed harvest dates define the management endpoint separately. Soil-water storage passes continuously through crops and fallows.

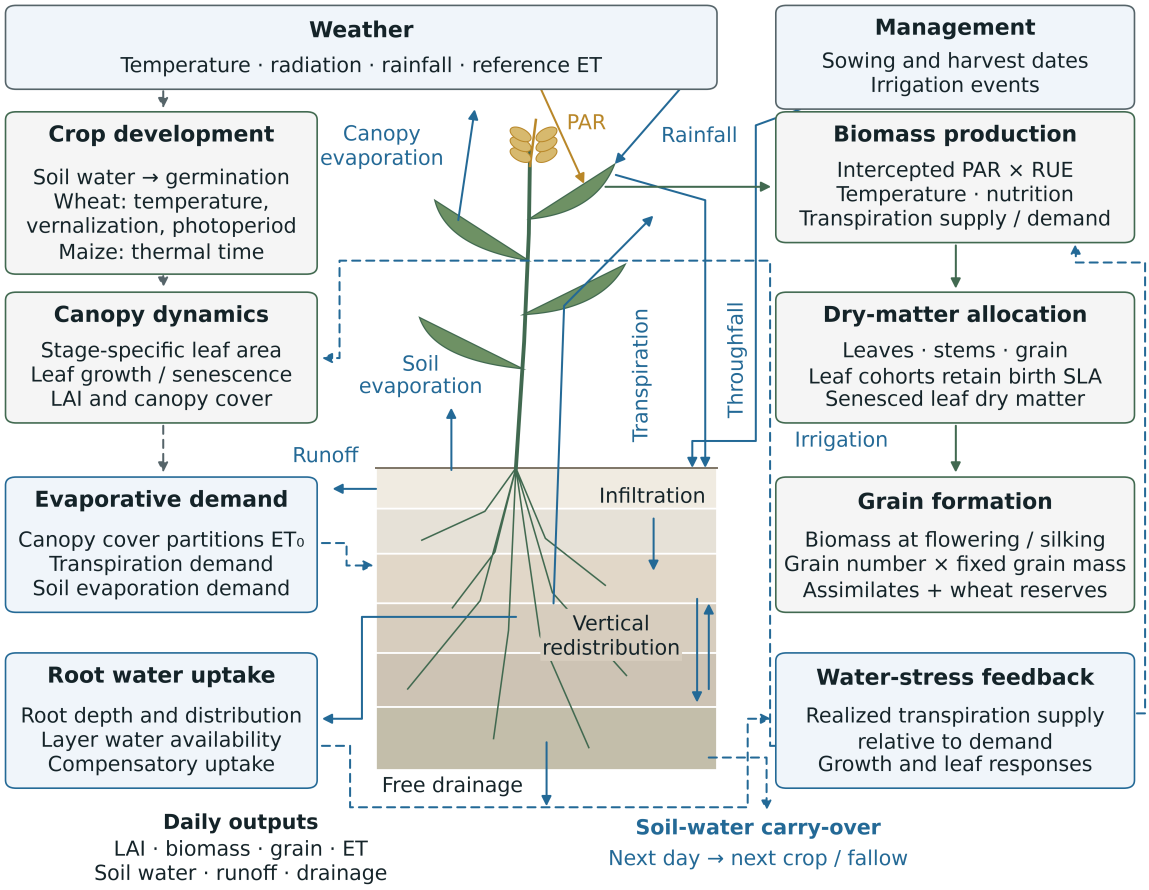



**Figure 4. Crop growth and soil-water coupling in Open Crop Model. Gold denotes photosynthetically active radiation (PAR) input; blue solid arrows denote water fluxes; green solid arrows denote dry-matter transfers; dashed arrows denote process controls and feedbacks. Nutrition is prescribed, the lower boundary uses free drainage, and soil water carries through crop/fallow segments. Plant and soil sketches are not to scale.**

Canopy interception of radiation follows Beer–Lambert attenuation. Aboveground production depends on intercepted PAR, crop-specific RUE, temperature, prescribed nutrition and an empirical power response to realized transpiration supply. PAR is 0.45 of incident shortwave radiation (Kiniry et al., 1989). Stage-specific SLA determines new leaf area, while established cohorts retain their birth area-to-mass ratio (de Wit and Berghuijs, n.d.). Stage-dependent thermal senescence and a separate daily stress-loss fraction determine leaf survival (Tan et al., 2020). LAI emerges from these carbon and area balances without a maximum-LAI parameter. Simulated biomass at wheat flowering or maize silking determines grain number; fixed individual grain mass defines sink capacity, and assimilate supply and wheat stem reserves constrain grain filling.

The water balance represents interception, ponding, infiltration, evaporation, root uptake, runoff and bottom drainage. Atmospheric demands are partitioned from reference ET using canopy cover and effective transpiration and evaporation coefficients. Soil evaporation follows a storage-based two-stage reduction (Allen et al., 1998). Root uptake depends on rooted-layer water, local depletion stress and a bounded compensation fraction that redirects unmet demand toward better-supplied rooted layers. Vertical transport uses van Genuchten–Mualem hydraulic functions (Mualem, 1976; van Genuchten, 1980), signed gravity–matric gradients and conservative storage bounds. Relative-arithmetic interface mobility and adaptive internal steps permit wetting and redistribution within 24 daily outer steps. The lower boundary is free drainage with no capillary-rise supply. This explicit layered approximation differs from a full implicit Richards-equation solution. Profile depths follow the supplied inputs, including 2 m for Wuqiao and regional simulations.

## 2.5. Model parameterization, calibration and evaluation

Phenological coefficients, stage-specific specific leaf area and senescence, initial LAI, stem-reserve coefficients and nutrition modifiers follow the crop parameterization described in Section S2. RUE, leaf allocation and grain-number parameters were estimated from 480 wheat and 433 maize growth and harvest observations across four wheat sites and five maize sites. Whole site-years retain their original partitions, with balanced contributions from variables, sites, site-years and crop cases. Station simulations retain their recorded site and cultivar parameterization; Wuqiao and regional simulations use the shared crop parameters (Table S3). Individual grain masses are fixed at 0.045 g for wheat and 0.300 g for maize.

Eight water-use parameters per crop are fitted to the documented Wuqiao treatments in 2016–2018: the transpiration coefficient, soil evaporation coefficient, readily available water fraction, assimilation water-stress exponent, root-density decay, maximum rooting depth, root water-uptake coefficient and root compensation fraction. Each crop contributes 12 seasonal ET, 12 dry-grain and 12 dry-biomass targets, with three annual W3−W0

ET contrasts derived from the same treatment seasons. RUE, leaf allocation and grain-number parameters remain fixed. Bounded least squares combines scaled observation residuals and weak parameter priors; objective weights, bounds and convergence settings are specified in Section S2.4.

The fitted water parameters are fixed before comparison with the four 2019 treatment means per crop. Previous inspection of these outcomes makes the test retrospective. Station observations have zero weight in this water fit and assess transfer under unconfirmed complete irrigation histories. Regional strategy selection uses historical weather responses; the crop-calibration years overlap the 2014–2025 regional comparison period, so those outcomes are conditional scenarios rather than time-independent crop validation. Observed and simulated quantities retain matched sampling windows and grain-moisture bases.

## 2.6. Regional upscaling and irrigation scenarios

Thirty-two area-weighted climate–soil representatives were selected from actual source cells using deterministic clustering of latitude, longitude, plant-available water, and 1997–2013 mean precipitation and reference ET. Features were standardized with rotation-area weights. Each cluster uses the actual cell nearest its centroid; the full source-cell mapping and represented areas are retained. Climate years used for policy evaluation do not enter the climate features used for selection. Shared crop parameter sets are transferred regionally without extrapolating site-specific adjustments.

Each representative has five continuous histories at irrigation fractions 0, 0.25, 0.50, 0.75 and 1.00. The conventional wheat schedule is 60 mm at sowing on 12 October, 90 mm on 25 March, 90 mm on 15 April and 60 mm on 10 May, totaling 300 mm. Maize receives 80 mm at sowing on 15 June. These common schedules are declared scenarios and are not observed universal practice across the plain. Every event amount is multiplied by the irrigation fraction, with its timing fixed. Wheat is simulated through 10 June and maize through 5 October; intervening fallows carry the soil state forward. Initial plant-available water is 80% at the start of the 1996 spin-up, with no subsequent seasonal storage reset. All crop and fallow ET, drainage and runoff are included. The final 2025 cycle ends on 5 October because the next fallow lies outside the declared endpoint. Nutrition is represented by prescribed fertilized-crop modifiers; nitrogen cycling and spatially varying nutrient limitations are not dynamically simulated.

Eight purposively selected non-representative cells provide paired spatial aggregation checks under all five irrigation fractions. Two cells from each latitude quartile have the largest area-times-feature-distance score. Actual-cell weather and profiles use the same continuous model and crop parameters. Their differences from

representative-cell predictions measure crop-model aggregation error rather than field-observation accuracy (Supplementary Section S3).

## 2.7. Irrigation strategies and comparisons

### 2.7.1. Spatial allocation under common irrigation budgets

Uniform allocation assigns the same irrigation fraction to every represented hectare. Targeted allocation chooses area shares $x_{jk}$ for representative j and simulated irrigation fraction k by linear programming. The objective maximizes $\sum_j A_j \sum_k x_{jk} \bar{Y}_{jk}$, where $\bar{Y}_{jk}$ is mean annual wheat-plus-maize grain in 1997–2013. Constraints require $\sum_k x_{jk} = 1$, $0 \le x_{jk} \le 1$ and $\sum_j A_j \sum_k x_{jk} I_k = (1 - r) \sum_j A_j I_1$, with reductions $r = 0.25$, 0.50 or 0.75. Within-representative mixtures describe shares of area receiving actual simulated irrigation treatments; crop responses are not interpolated between irrigation fractions. Allocations remain fixed in 2014–2025. Both policies deliver exactly the same total field-irrigation water in every year, so their differences reflect allocation under the same volume constraint.

Spatial contrasts at the common 50% irrigation cut use frozen shares of complete continuous irrigation histories. Their area-weighted grain, ET and drainage differences map from 32 representatives to 3,641 source cells. Fixed southern (32–36°N), central (36–38°N) and northern (38–41°N) bands summarize the distribution. Counts of positive annual contrasts describe consistency within the modeled record (Figure 5; Table S9).

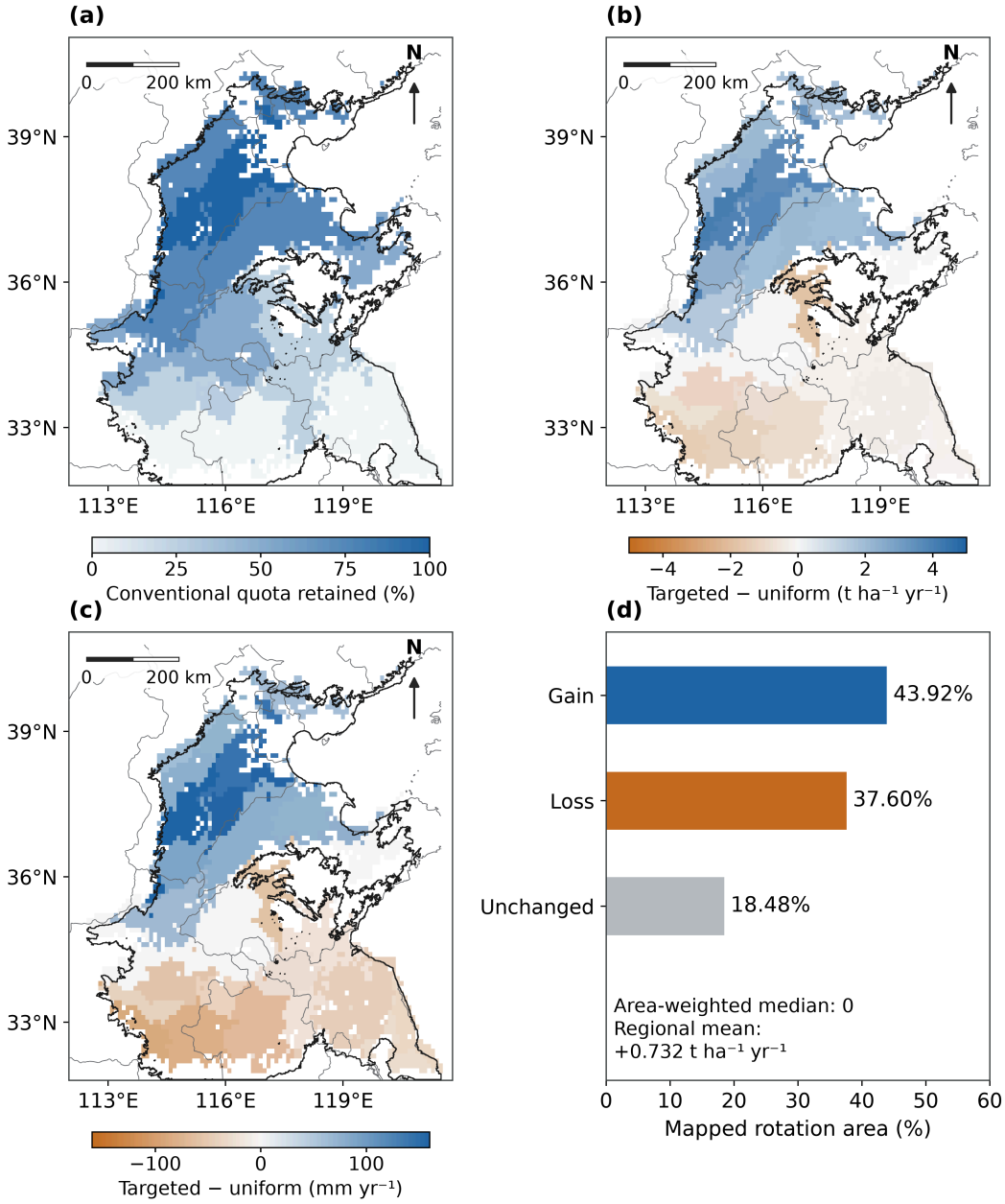



**Figure 5. Spatial redistribution at a common 50% regional irrigation reduction, 2014–2025. (a) Targeted fraction of the conventional quota; uniform allocation retains 50% everywhere. (b,c) Targeted-minus-uniform combined rotation dry grain and crop-plus-fallow ET. (d) Mapped rotation-area shares with positive, negative or unchanged mean grain contrasts. The 3,641 cells inherit 32 representative responses. Thin grey provincial lines and black study-domain outlines follow Figure 2; unmapped areas remain blank. North arrows indicate geographic north; scale bars use WGS84 ellipsoidal ground distances at their displayed latitude.**

## 2.7.2. Annual quotas based on antecedent water availability

Relative water-availability classes combine two antecedent indicators: regional total-water-storage status and local rainfall before wheat sowing. The storage indicator is the mean of available August and September GRACE observations in the calendar year preceding harvest, with at least one monthly solution required. Precipitation is

accumulated from 1 July to 30 September at each simulation unit, covering 92 days before the prescribed 12 October wheat sowing. The corresponding precipitation/reference-ET ratio is retained as a diagnostic of climatic water availability. These antecedent indicators do not use rainfall or crop outcomes from the subsequent growing season.

Thresholds use 2003–2013 harvest years, the common selection period with antecedent GRACE coverage. The regional storage median is 13.38 mm, and the rotation-area-weighted precipitation median is 431.74 mm. Values below their respective thresholds define lower storage and lower rainfall; values at or above define higher storage and higher rainfall. Crossing the two indicators gives four hydroclimatic classes. The classes describe relative antecedent water availability, rather than absolute water scarcity derived from water demand, allocation rights or access. Spatial differences within a year arise from precipitation; GRACE does not supply independently resolved field-scale storage classes.

For each class, five continuous irrigation histories provide paired grain and ET responses relative to conventional irrigation, weighted by rotation area and year. The primary candidate is the lowest fraction retaining at least 95% of class-mean conventional grain in 2003–2013; a 98% target assesses a stricter production criterion. Selected fractions and class thresholds remain frozen in 2014–2025. Higher-storage classes have no evaluation exposure. Required GRACE observations are missing for 2014, 2018 and 2019.

Two annual strategies apply the irrigation fractions selected from fixed-fraction responses at grain-retention targets of 95% and 98%. At each simulation unit, the antecedent storage–rainfall class determines the fraction before wheat sowing. This fraction scales all four wheat applications and the subsequent maize application in the same rotation year. Application dates, crop parameters, weather, hydraulic profiles and the free-drainage lower boundary are identical to those in the fixed-fraction experiment. Conventional irrigation applies before 2003 and whenever an eligible storage class is unavailable, including evaluation harvest years 2014, 2018 and 2019.

Each strategy is simulated continuously from the 1996 spin-up through 2025. Soil-water states pass between crops and fallows and across annual changes in irrigation fraction. Conventional irrigation supplies the paired continuous reference; replays at two simulation units verify the scenario driver against conventional outputs. Class thresholds and irrigation fractions remain fixed after selection. Outcomes cover all twelve comparison years and, separately, the nine years with eligible classes. Production retention is the ratio of period-mean regional grain production to its corresponding conventional mean; annual ratios describe variation within the period. The strategies assume access to antecedent observation periods, while release availability of GRACE before sowing remains unverified.

### 2.7.3. Rainfall-only controls and matched monitoring coverage

Rainfall-only controls use the same 2003–2013 simulation-unit years with eligible GRACE observations, the same rotation-area weights, irrigation candidates and grain-retention targets. Selection groups responses by antecedent rainfall alone and chooses the lowest fraction meeting the relevant target. The rainfall threshold is identical to that in the joint classification. Matched controls receive conventional irrigation whenever the storage–rainfall strategy lacks an eligible class, preserving both the selection cohort and monitoring coverage. Continuous differences between these paired strategies consequently assess the incremental effect of storage-based grouping.

Separate rainfall-available controls apply the selected rainfall-only fractions whenever an eligible rainfall group exists, including years without GRACE coverage. Their complete histories assess the combined effect of rainfall-based management and greater monitoring availability. Irrigation changes in earlier gap years can alter the soil-water state of later crops, so these controls are reported separately from the matched comparison.

## 2.8. Evaluation metrics, spatial statistics and water accounting

Performance is summarized by RMSE, mean bias error, nRMSE relative to observed mean, Nash–Sutcliffe efficiency (NSE) (Nash and Sutcliffe, 1970) and squared Pearson correlation ($R^2$). Record-level station scores give equal weight to available sites, site-years within sites, crop cases within site-years and records within cases. Wuqiao treatment means receive equal weights within each crop and partition. Field calibration and retrospective testing remain separate. Grain and harvest biomass in station comparisons are dry matter; Wuqiao grain results retain the published 13% moisture basis, with a single 0.87 conversion between dry and reported grain. Biomass remains dry matter throughout. $R^2$ describes association and can remain high despite systematic bias; agreement with observations is assessed jointly with RMSE, bias and NSE. Seasonal-curve annotations use paired plotted bin means with equal weights across bins and counts of paired bins. Treatment-season scatter annotations use individual treatment means. Each partition is scored separately in the displayed variable units.

Seasonal curves group matching observations and predictions into 15-day wheat and 7-day maize bins at observation-window midpoints in days after sowing. Case means receive equal weight within site-years, site-year means within sites and available sites within bins. Standard errors describe variation among contributing site-year means rather than model uncertainty. Site-specific curves use equal site-year weights. Coverage varies among bins, and gaps remain uninterpolated. Record-level accuracy scores use unbinned records; seasonal-curve annotations describe the paired bin means. Yucheng harvest error bars describe quadrat sampling variation within crop-years.

ET sums use identical observed and simulated measurement days within each case and retain the station balancing weights. They exclude unmeasured days and do not represent complete-season observed totals. Regional grain production uses dry mass and rotation-area weights; actual ET includes crops and fallows. Paired policy differences cover 2014–2025. Daily water and carbon residuals and continuity of carried soil-water storage assess numerical conservation. Field irrigation, ET, drainage below 2 m and groundwater storage have distinct accounting boundaries.

Spatial distribution statistics describe targeted-minus-uniform policy contrasts in mean annual combined wheat–maize dry grain and water fluxes over 2014–2025. The 3,641 mapped 0.1° source cells carry outcomes from 32 representative response units. Cell weights are the static 2020 mapped rotation areas after cropland-fraction correction; they represent mapped rotation area rather than observed harvested area. Gains and losses exceed $+10^{-6}$ and fall below $-10^{-6}$ kg ha$^{-1}$, respectively; contrasts within these bounds are classified as unchanged. Group means use rotation-area weights, regional contributions sum area multiplied by the corresponding contrast, and weighted quantiles use the inverse empirical cumulative area distribution. Annual loss frequencies use the same sign threshold across the 12 years. Connected loss footprints comprise occupied cells sharing a horizontal or vertical grid edge (rook adjacency). The statistics describe the conditional modeled allocation footprint; mapped cells do not provide independent replication or cell-specific validation.

# 3. Results

## 3.1. Conditional multisite crop comparisons

Retrospective-evaluation LAI RMSE was 1.843 m² m$^{-2}$ for wheat and 0.995 m² m$^{-2}$ for maize, with NSE of 0.184 and 0.689, respectively (Figure 6). LAI RMSE in the original calibration partition was 1.727 and 1.138 m² m$^{-2}$. In-season biomass RMSE in retrospective evaluation was 4.413 and 2.789 t ha$^{-1}$, with biases of +1.314 and +1.158 t ha$^{-1}$ and NSE of 0.349 and 0.796. Site-specific seasonal comparisons retain their available observations and original partitions (Figures S6–S11). Record-level performance scores use unbinned observations with hierarchical weights; the seasonal-curve annotations quantify agreement between the plotted bin means.

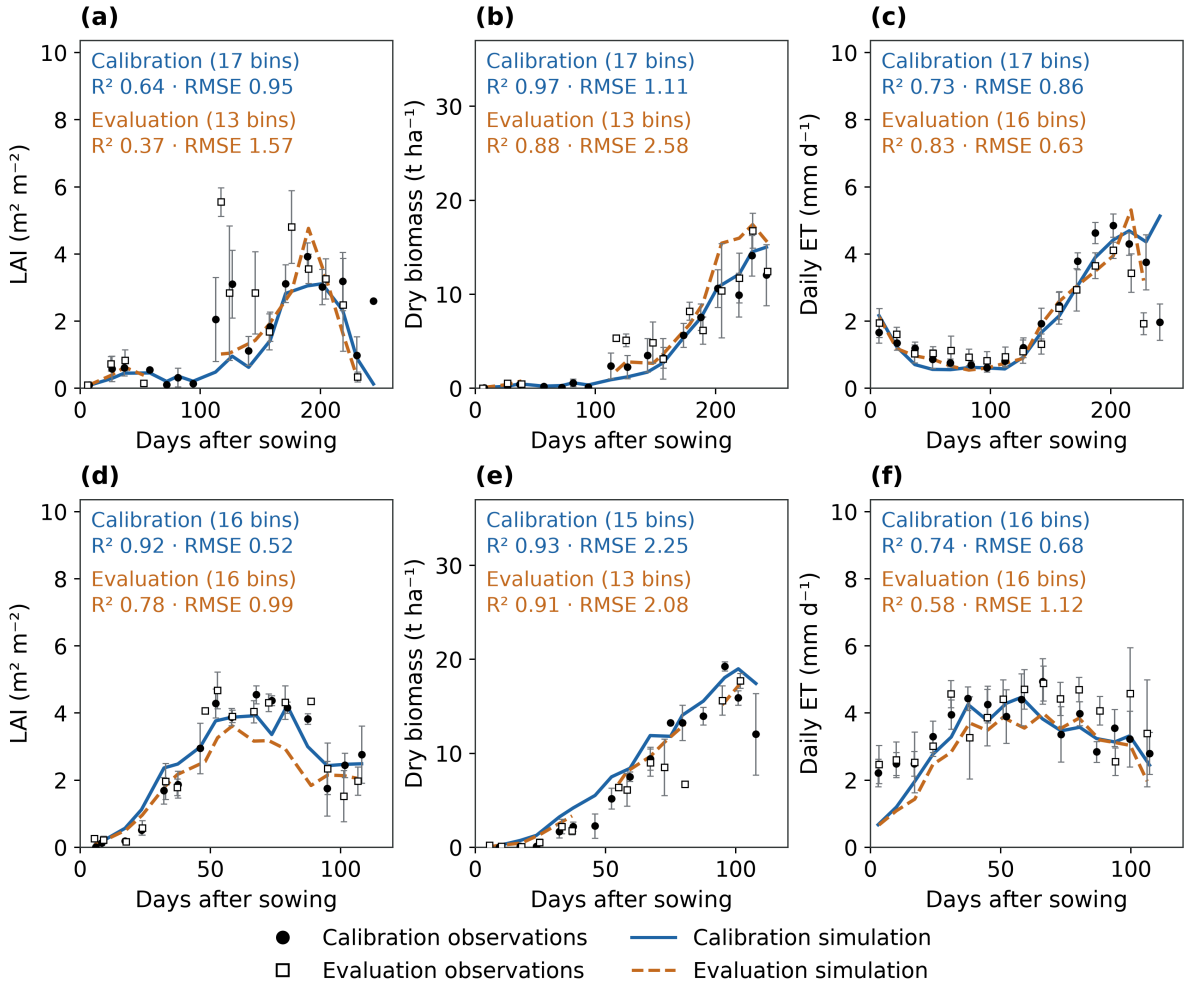



**Figure 6. Multisite seasonal crop comparisons. Wheat and maize occupy the top and bottom rows; columns show LAI, aboveground dry biomass and daily actual ET. Filled and open observations distinguish the original calibration and retrospective-evaluation partitions; solid blue and dashed orange lines show the corresponding simulations. Error bars describe standard errors among site-year means. Balanced means use 15-day wheat and seven-day maize bins, and unobserved bins remain gaps. Complete station irrigation management remains unconfirmed. Colored annotations give calibration and retrospective-evaluation $R^2$ and RMSE for the plotted seasonal means. Each paired bin has equal weight, and counts refer to bins rather than individual observations or independent growing seasons. $R^2$ is squared Pearson correlation; RMSE uses the corresponding vertical-axis units. Record-level accuracy is evaluated separately.**

Dry-grain RMSE was 1.256 t $ha^{-1}$ for wheat and 1.402 t $ha^{-1}$ for maize in the original calibration partition, compared with 0.834 and 2.286 t $ha^{-1}$ in retrospective evaluation (Figure S5). Evaluation grain biases were +0.485 and -1.586 t $ha^{-1}$, and NSE was 0.099 and -1.566. Evaluation harvest-biomass RMSE was 2.736 t $ha^{-1}$ for wheat and 3.719 t $ha^{-1}$ for maize; the corresponding observation counts were 54 in nine wheat years and 48 in eight maize years.

Daily ET RMSE in retrospective evaluation was 1.809 mm $d^{-1}$ for wheat and 2.572 mm $d^{-1}$ for maize, with biases of +0.141 and -0.734 mm $d^{-1}$ and NSE of 0.121 and 0.012 (Figure 6). ET summed over matched measurement days had evaluation RMSE of 51.29 mm for wheat and 74.17 mm for maize, biases of +25.94 and −55.58 mm, and NSE of 0.826 and 0.118 (Table 2). These sums exclude days without ET measurements.

**Table 2. Conditional station ET sums on matched measurement days. Observed and simulated sums use identical measurement days within cases, with balanced site, site-year and case weights. Unmeasured days are excluded. Original calibration partitions retain an earlier growth-fitting history; evaluation is retrospective. Complete irrigation-log matching remains unconfirmed.**

| Crop | Dataset | Seasons | NSE | RMSE (mm) | MBE (mm) |
|---|---|---|---|---|---|
| Maize | Prior calibration partition | 11 | 0.511 | 45.60 | -34.92 |
| Maize | Retrospective evaluation | 10 | 0.118 | 74.17 | -55.58 |
| Wheat | Prior calibration partition | 11 | -0.300 | 84.94 | -23.53 |
| Wheat | Retrospective evaluation | 11 | 0.826 | 51.29 | 25.94 |

## 3.2. Wuqiao calibration and retrospective testing

Seasonal ET calibration RMSE was 40.19 mm for wheat and 37.44 mm for maize across 12 treatment seasons per crop, with biases of −12.65 and +3.60 mm and nRMSE of 8.97% and 12.69% (Table 3; Figure 7). Across the four 2019 testing means per crop, ET RMSE was 73.80 and 55.64 mm, bias was +48.93 and −35.90 mm, nRMSE was 21.98% and 14.13%, and NSE was 0.503 and −0.346, respectively.

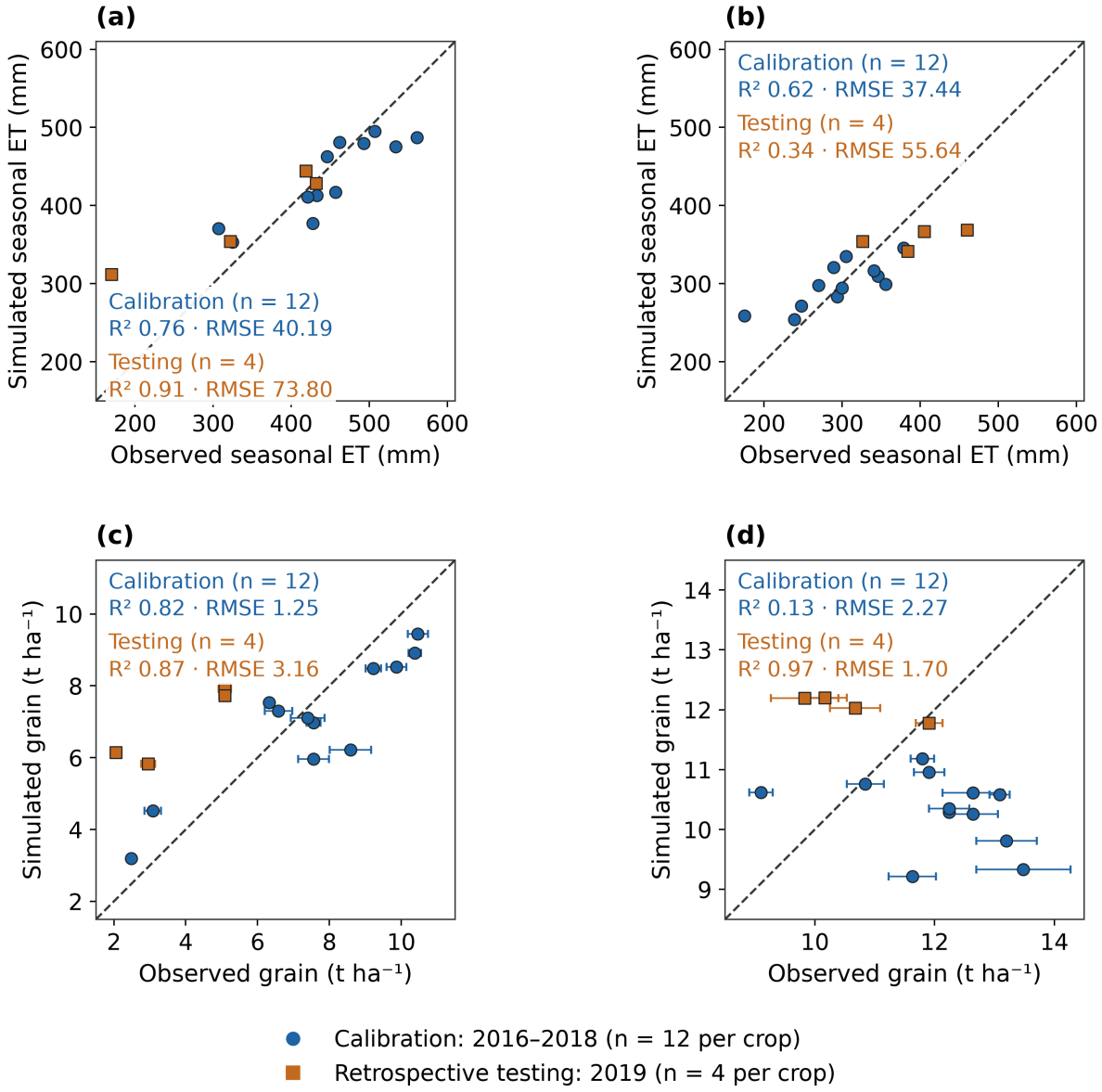



**Figure 7. Wuqiao calibration and retrospective testing. (a,b) Seasonal ET; (c,d) annual treatment-mean grain at 13% moisture. Wheat occupies the left column and maize the right. Circles show 2016–2018 treatments used to fit water-use parameters; squares show 2019, withheld from the fit and selection. Dashed lines denote 1:1 agreement. Crop-specific grain axes focus on their observed ranges. Horizontal grain error bars are the published SE from three field replicates. Observed ET follows the published 0–2 m water balance; simulated dry grain is converted once by division by 0.87. Maize labels retain preceding wheat management with identical maize irrigation within each year. Colored annotations give separate calibration and retrospective-testing $R^2$, RMSE and n, using equally weighted treatment-season means. $R^2$ is squared Pearson correlation and measures association rather than agreement with the identity line; RMSE is in the axis units.**

At 13% grain moisture, calibration RMSE was 1.249 t ha$^{-1}$ for wheat and 2.267 t ha$^{-1}$ for maize; retrospective-testing RMSE was 3.161 and 1.700 t ha$^{-1}$. Testing grain biases were +3.108 and +1.403 t ha$^{-1}$, with nRMSE of 83.27% and 15.97% and NSE of −4.654 and −3.641 (Table 3; Figure 7). Dry aboveground-biomass RMSE was

4.222 and 4.688 t ha$^{-1}$ in calibration and 5.427 and 2.038 t ha$^{-1}$ in testing, respectively. Testing biomass nRMSE was 55.66% for wheat and 10.44% for maize.

W3−W0 seasonal ET-contrast RMSE was 45.79 mm for wheat and 66.84 mm for maize across three calibration years; the absolute 2019 contrast errors were 116.37 and 52.74 mm. For grain contrasts of W1–W3 relative to same-year W0, calibration RMSE was 0.519 and 1.669 t ha$^{-1}$ and retrospective-testing RMSE was 1.316 and 1.323 t ha$^{-1}$, respectively, at 13% moisture (Table S5; Figure S2). Maize treatment labels denote preceding wheat irrigation; maize irrigation was identical within each year.

**Table 3. Wuqiao bias, relative error and efficiency by fitting and testing partition.**

| Crop | Period | Target | n | Unit | MBE | nRMSE (%) | NSE |
|---|---|---|---|---|---|---|---|
| Maize | 2016–2018 | Seasonal ET | 12 | mm | 3.604 | 12.69 | 0.533 |
| Maize | 2016–2018 | Grain | 12 | t ha$^{-1}$ | -1.738 | 18.78 | -2.940 |
| Maize | 2016–2018 | Biomass | 12 | t ha$^{-1}$ | -4.188 | 21.79 | -5.625 |
| Maize | 2019 | Seasonal ET | 4 | mm | -35.902 | 14.13 | -0.346 |
| Maize | 2019 | Grain | 4 | t ha$^{-1}$ | 1.403 | 15.97 | -3.641 |
| Maize | 2019 | Biomass | 4 | t ha$^{-1}$ | 0.726 | 10.44 | -0.313 |
| Wheat | 2016–2018 | Seasonal ET | 12 | mm | -12.649 | 8.97 | 0.689 |
| Wheat | 2016–2018 | Grain | 12 | t ha$^{-1}$ | -0.445 | 16.74 | 0.746 |
| Wheat | 2016–2018 | Biomass | 12 | t ha$^{-1}$ | -2.480 | 25.48 | 0.496 |
| Wheat | 2019 | Seasonal ET | 4 | mm | 48.929 | 21.98 | 0.503 |
| Wheat | 2019 | Grain | 4 | t ha$^{-1}$ | 3.108 | 83.27 | -4.654 |
| Wheat | 2019 | Biomass | 4 | t ha$^{-1}$ | 4.908 | 55.66 | -0.752 |

Calibration uses 12 treatment seasons per crop in 2016–2018; retrospective testing uses four in 2019. Counts are treatment means rather than independent replicate seasons. ET is a 0–2 m water-balance estimate under negligible runoff and drainage assumptions. Grain is at 13% moisture and aboveground biomass is dry matter. Annual grain means and SE are digitized from source Supplementary Figure S2. Initial total storage is observed; layer distribution and transferred hydraulics remain conditional.

## 3.3. Grain production and water balance under irrigation allocation

At a 50% regional irrigation reduction, targeted allocation gave a simulated annual rotation dry-grain yield of 15.761 t ha$^{-1}$ yr$^{-1}$ for wheat and maize combined, compared with 15.029 under uniform cuts. Both yields remained below the conventional yield of 16.756 t ha$^{-1}$ yr$^{-1}$. Targeting therefore retained 0.732 t ha$^{-1}$ yr$^{-1}$ more grain than uniform allocation (4.87% of uniform production), equivalent to 7.219 million t yr$^{-1}$ across the fixed 9.867 million rotation hectares (Table 4; Figure 8). Targeted production remained 9.813 million t yr$^{-1}$ below conventional production.

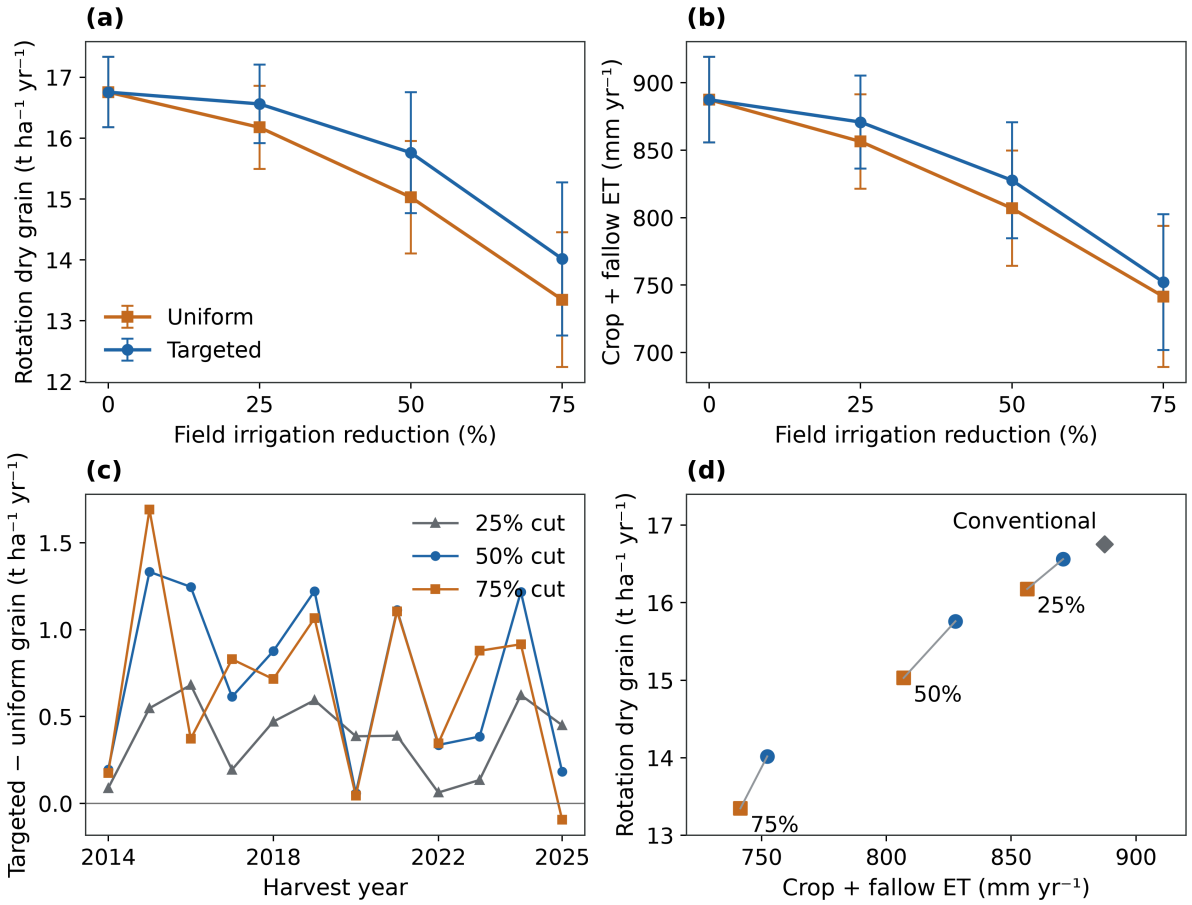



**Figure 8. Irrigation allocation and grain–water trade-offs during 2014–2025. (a,b) Rotation-area-weighted annual combined wheat–maize dry grain and crop-plus-fallow ET; error bars are between-year SD. (c) Annual targeted-minus-uniform grain at identical irrigation budgets. (d) Period means; connectors join policies with the same budget. Values per hectare use the same 9.867 million mapped rotation hectares, rather than separate crop areas. Targeting retains more grain than uniform cuts while remaining below conventional production.**

Both allocations supplied 190 mm yr$^{-1}$ of field irrigation. Targeted and uniform crop-plus-fallow ET averaged 827.70 and 806.94 mm yr$^{-1}$, respectively. The grain advantage was accompanied by 20.76 mm yr$^{-1}$ (2.048 km$^3$ yr$^{-1}$) greater ET. Profile drainage was 79.04 versus 96.34 mm yr$^{-1}$, and runoff was 14.43 versus 18.99 mm yr$^{-1}$ (Table 4).

The targeted-minus-uniform grain contrast at the 50% cut was positive in all twelve comparison years (Figure 8c). The positive regional contrast was accompanied by spatially contrasting yield responses (Section 3.4).

**Table 4. Simulated annual rotation production and water balance under equal irrigation reductions, 2014–2025.**

| Strategy | Dry grain (t ha$^{-1}$ yr$^{-1}$) | Irrigation (mm yr$^{-1}$) | ET (mm yr$^{-1}$) | Drainage (mm yr$^{-1}$) | Runoff (mm yr$^{-1}$) |
|---|---|---|---|---|---|
| Conventional | 16.756 | 380.00 | 887.43 | 185.26 | 39.32 |
| Uniform, 25% cut | 16.177 | 285.00 | 856.41 | 135.94 | 25.71 |
| Targeted, 25% cut | 16.562 | 285.00 | 870.81 | 125.90 | 19.86 |
| Uniform, 50% cut | 15.029 | 190.00 | 806.94 | 96.34 | 18.99 |
| Targeted, 50% cut | 15.761 | 190.00 | 827.70 | 79.04 | 14.43 |
| Uniform, 75% cut | 13.347 | 95.00 | 741.57 | 68.72 | 15.83 |
| Targeted, 75% cut | 14.018 | 95.00 | 752.30 | 60.01 | 13.68 |

All per-hectare means use the same 9.866602 million mapped rotation hectares. Dry grain combines wheat and maize harvests; ET includes crops and fallows. Regional grain in million tonnes equals rotation yield multiplied by mapped area in million hectares. Regional water in km³ equals depth in mm multiplied by hectares and 10, divided by $10^9$. The conventional schedule supplies 300 mm to wheat and 80 mm to maize.

## 3.4. Spatial distribution and persistence of yield contrasts

Relative to uniform allocation, the central and northern latitude bands contributed grain gains of 7.475 and 1.886 million t yr$^{-1}$, while the southern band contributed a loss of 2.142 million t yr$^{-1}$. The southern band occupied 60.7% of mapped rotation area. Mean rotation-yield contrasts were +2.415, +2.420 and −0.357 t ha$^{-1}$ yr$^{-1}$ in the central, northern and southern bands, respectively (Figure 5; Table S9).

At the 50% regional irrigation reduction, targeted allocation increased mean annual combined wheat–maize dry grain relative to uniform reduction on 43.92% of the mapped rotation area (4.333 Mha), decreased it on 37.60% (3.710 Mha), and left it unchanged on 18.48% (1.823 Mha). Mean contrasts within the gain and loss areas were +2.439 and -0.903 t ha$^{-1}$ yr$^{-1}$, respectively. Their regional contributions were +10.568 and -3.349 Mt yr$^{-1}$, giving a net regional grain difference of 7.219 Mt yr$^{-1}$ (0.732 t ha$^{-1}$ yr$^{-1}$ across the full mapped area). Local losses offset 31.69% of the gross positive contribution (Figure 5d).

The area-weighted median grain contrast was 0.000 t $ha^{-1}$ $yr^{-1}$, and the 10th and 90th percentiles were -1.182 and +3.553 t $ha^{-1}$ $yr^{-1}$. Relative to uniform reduction, mean annual field irrigation increased by 119.0 mm $yr^{-1}$ in the gain area and decreased by 139.0 mm $yr^{-1}$ in the loss area. The corresponding crop-plus-fallow ET contrasts were +94.4 and -55.1 mm $yr^{-1}$. Irrigation changes balanced at regional scale, whereas the area-weighted mean ET contrast was +20.8 mm $yr^{-1}$.

Negative yield contrasts persisted over 2014–2025: 21.34% of the mapped rotation area had lower yields than under uniform allocation in all 12 years, and 37.60% experienced a loss in at least one year. Every response unit with a negative period-mean contrast recorded losses in at least 9 years; all response units with a positive period-mean contrast recorded gains in all 12 years. Across years, the mean annual area exposed to a loss was 35.37%. Mapped cells with negative period-mean contrasts formed 8 connected components under rook adjacency. The largest contained 3.700 Mha of mapped rotation area (99.73% of all loss area), with occupied cell centers extending from 32.0 to 36.3°N and 112.8 to 121.5°E.

Negative period-mean yield contrasts occupied 32.80%, 37.60%, and 60.08% of mapped rotation area at reductions of 25%, 50%, and 75%, respectively. At the 75% reduction, the area-weighted median contrast became negative (-0.788 t $ha^{-1}$ $yr^{-1}$), although gains on 37.04% of the area produced a positive regional balance. Gross positive and negative contributions at that reduction were +14.100 and -7.487 Mt $yr^{-1}$, leaving a net gain of 6.614 Mt $yr^{-1}$, below the net gain at the 50% reduction.

## 3.5. Antecedent water availability and annual irrigation strategies

Four antecedent storage–rainfall classes occurred during 2003–2013. Nine comparison years had eligible GRACE observations, all in the lower-storage category; three years without eligible GRACE observations received conventional irrigation quotas. At the 95% grain target, both storage strata selected an irrigation fraction of 0.75 under lower rainfall and 0.50 under higher rainfall. The 98% target selected 1.00 and 0.75, respectively (Table 5).

Continuous strategies selected at the 95% and 98% targets retained 96.27% and 99.63% of conventional grain production over the nine years with eligible GRACE observations. Field-irrigation reductions were 130.72 and 35.72 mm per rotation; corresponding ET reductions were 36.84 and 6.17 mm (Figure 9; Table S7). The 95% strategy retained at least 95% in 8 of nine years, with a minimum of 92.12%.

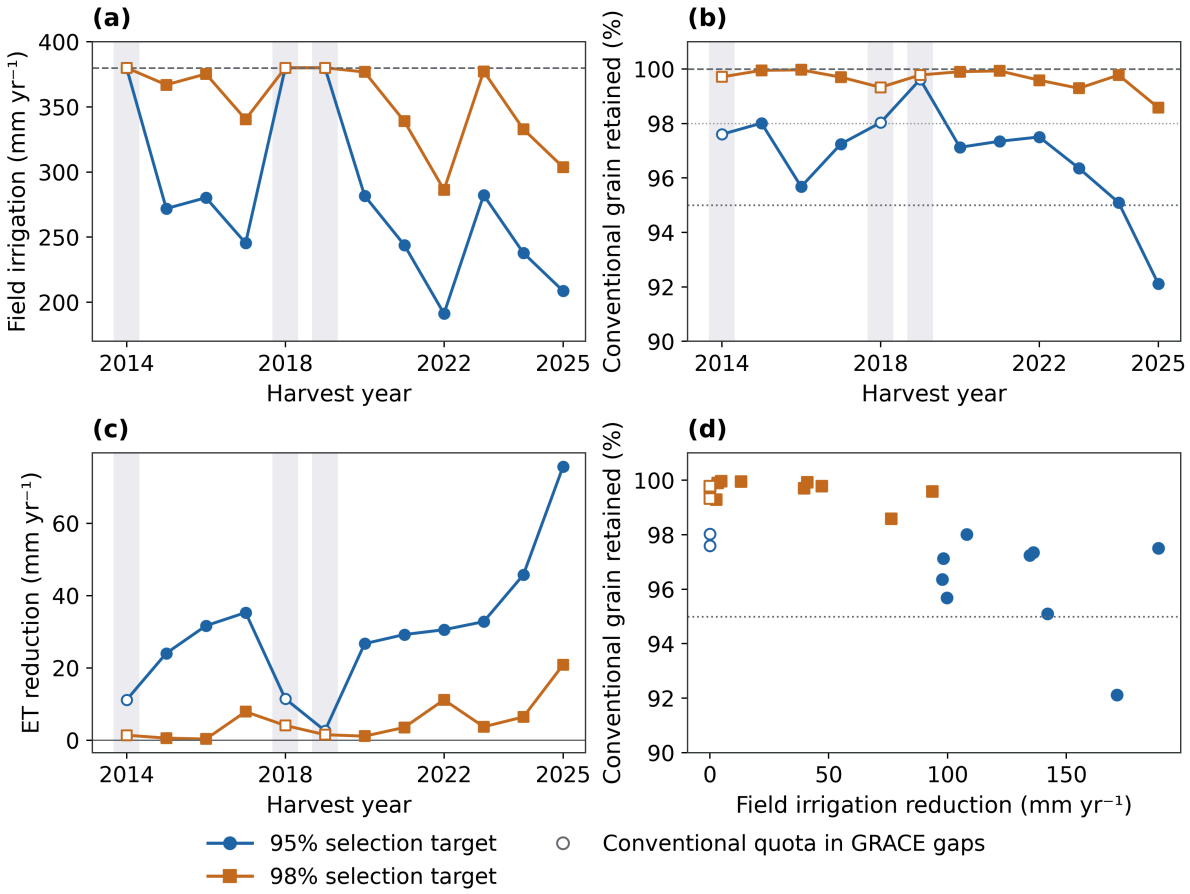



**Figure 9. Continuous annual irrigation strategies during 2014–2025. (a) Field irrigation; (b) grain retained relative to conventional irrigation; (c) crop-plus-fallow ET reduction; (d) annual production–irrigation combinations. Blue circles and orange squares identify 95% and 98% selection targets. Shading and open markers identify missing-GRACE years with conventional quotas. Grain and ET can still differ in those years because preceding strategy decisions alter carried soil-water states. Matched rainfall-only histories coincide with the storage–rainfall histories.**

Storage–rainfall and matched rainfall-only strategies selected identical irrigation fractions and produced identical grain, ET, drainage and irrigation histories at both targets. Rainfall-only strategies that applied decisions during GRACE gaps followed different irrigation histories and carried different soil-water states into subsequent crops (Table S8).

**Table 5. Selected irrigation candidates by relative water-availability class.**

| Class | Irrigation fraction | Wheat irrigation (mm) | Maize irrigation (mm) | Selection grain retention (%) |
|---|---|---|---|---|
| Higher storage dry | 0.750 | 225.000 | 60.000 | 95.537 |
| Higher storage wet | 0.500 | 150.000 | 40.000 | 97.093 |
| Lower storage dry | 0.750 | 225.000 | 60.000 | 96.617 |
| Lower storage wet | 0.500 | 150.000 | 40.000 | 97.431 |

Fractions meet the 95% class-mean grain target during selection years 2003–2013. Current shared crop parameters apply throughout. Both storage strata select identical fractions within each rainfall group; higher-storage classes have no later-period exposure.

# 4. Discussion

We used a calibrated crop model to simulate continuous winter wheat–summer maize rotations across the North China Plain, compare uniform and spatially targeted irrigation under equal water budgets, and test whether antecedent regional water storage improved decisions based on rainfall. Sustaining grain production under restricted irrigation requires consideration of both consumptive water use and the distribution of production losses across locations. The conditional results show that higher regional grain retention can accompany greater evapotranspiration and lower production in some locations, whereas storage information did not change the selected irrigation quotas within the tested design. The management value of these strategies therefore depends on accounting for the complete rotation water balance, soil-water carry-over across crops and fallows, and reliable crop responses to irrigation.

## 4.1. Marginal irrigation returns and spatial allocation

Spatial targeting exploits differences in marginal grain response to irrigation. Diminishing production gains across field schedules (Yang et al., 2022) and spatial variation in irrigation requirements (Wu et al., 2019) provide a basis for redistributing a limited supply, consistent with regional allocation studies (Jiang et al., 2016). The equal-budget comparison attributes the simulated production difference to delivery allocation. Complete continuous histories retain the effects of earlier irrigation and rainfall on subsequent crop responses.

At the 50% delivery reduction, targeting retained 0.732 t ha$^{-1}$ yr$^{-1}$ more combined wheat–maize dry grain than uniform cuts. Across 9.8666 million rotation hectares, the corresponding reduction in grain loss was 7.219 million t yr$^{-1}$, or 4.87% of the uniform total. Targeted production remained below conventional production. The positive contrast in all twelve years indicates consistent regional gains within the simulated record. Its magnitude remains dependent on the crop-response functions and declared management schedules (Table 4; Figure 8).

The spatial distribution sharpens this distinction between regional and local outcomes. Relative to uniform allocation, mean losses occupy 37.60% of mapped rotation area, and losses persist in every comparison year on 21.34%. Gross gains of 10.568 Mt yr$^{-1}$ are partly offset by losses of 3.349 Mt yr$^{-1}$; the area-weighted median is zero despite a positive regional mean. The regional grain objective balances gains and losses among locations. Local production floors constitute an additional allocation constraint.

Regional gain has a pronounced distributional cost. Central and northern band-mean rotation yields increased by 2.415 and 2.420 t ha$^{-1}$ yr$^{-1}$, while the southern mean decreased by 0.357 t ha$^{-1}$ yr$^{-1}$. The southern band covers 60.7% of the rotation area, so its losses substantially offset gains elsewhere (Figure 5; Table S9). The production objective maximizes the area-weighted regional total and supplies no minimum yield for an individual location. Local production floors and constraints on the distribution of irrigation reductions are therefore separate requirements for allocation that protects affected farms.

## 4.2. Rotation carry-over and the timing of water supply

The spatial production response integrates water effects across both crops. Wheat irrigation can support canopy growth and grain formation, replenish water available to maize or increase drainage. Reduced wheat delivery can increase subsequent dependence on summer rainfall, producing different annual grain responses under different rainfall sequences. The sixteen-year irrigation experiment shows diminishing production gains, and the Wuqiao experiment links precipitation conditions with wheat scheduling and subsequent maize production (Yang et al., 2024; Yang et al., 2022). These coupled responses make the complete rotation the relevant unit for irrigation allocation.

Continuous soil-water histories preserve the delayed benefits and costs of allocation and annual quota changes (Figure S3). Production can be maintained by drawing down profile storage, with later rainfall or irrigation determining its replenishment. A finite-period delivery reduction can therefore include a contribution from stored

soil water. Crop-plus-fallow accounting also captures evaporation and drainage between harvest and sowing. Their inclusion connects apparent production retention with the water remaining for the next crop and the complete rotation water balance.

The field evidence constrains only part of this carry-over mechanism. Wuqiao simulations use reported total profile storage with an assumed layer distribution. Wheat crop-start storage follows the pre-sowing simulation, whereas maize is independently initialized from treatment-specific observations. Maize irrigation is identical among treatments within each year, and its treatment labels identify preceding wheat schedules (Yang et al., 2024). Maize comparisons thus evaluate crop responses to the initial state associated with wheat management (Figure 7; Figure S2). Regional carry-over additionally depends on the model's continuous propagation and vertical redistribution of water between crops.

## 4.3. Production, consumptive use and groundwater accounting

The production benefit of targeting carries a consumptive-use trade-off. At the common 50% delivery reduction, targeting increased ET by 2.048 $km^3\ yr^{-1}$ and reduced profile drainage by 1.707 $km^3\ yr^{-1}$ relative to uniform allocation. With rainfall and total irrigation held constant, differences in ET are balanced by drainage, runoff and soil-storage change (Table 4). A shared delivery constraint therefore permits different consumptive outcomes. Irrigation productivity and consumptive-water savings require distinct objectives (Grafton et al., 2018; Perry et al., 2009).

The soil-profile balance also has a different boundary from the aquifer balance. Drainage below 2 m enters a deeper vadose zone, with storage and transit outside the modeled profile (Wu et al., 2023). Reduced field delivery or profile drainage therefore provides insufficient information to quantify changes in pumping, aquifer recharge or groundwater storage. Production and ET comparisons can identify agronomic trade-offs, while groundwater outcomes require the corresponding subsurface and abstraction accounting.

GRACE offers regional storage context at a broader scale. Its anomalies integrate groundwater with other terrestrial stores, and its effective observation scale exceeds the displayed grid resolution (Save et al., 2016; Scanlon et al., 2016). Recent aquifer recovery reflects water transfers, abstraction management and hydroclimatic conditions (Long et al., 2025; Xu et al., 2026). The crop-model scenarios quantify changes in field delivery, consumption and profile outflow without attributing those observed storage changes to individual irrigation strategies.

## 4.4. When storage information changes an irrigation decision

A storage indicator adds decision value when its inclusion changes the selected action or improves the grain–water outcome under common constraints. GRACE represents regional storage conditions, while antecedent rainfall describes more local water input (Save et al., 2016; Scanlon et al., 2016). Their different physical support motivates joint classification, but decision value also depends on how the resulting classes distinguish among the available irrigation fractions.

Both storage strata selected the same fraction for a given rainfall group at the 95% and 98% grain targets. Storage–rainfall and matched rainfall-only strategies consequently produced identical complete histories of irrigation, crop states, grain, ET and drainage. The storage split supplied additional environmental description without changing an irrigation action in this five-level quota design. All nine later years with eligible observations fell in the lower-storage category, so the higher-storage choices also lacked later-period exposure (Table 5; Figure 9). These features delimit the configuration and conditions represented by the equal outcomes.

Monitoring coverage must remain distinct from the grouping criterion. Rainfall-available controls can change irrigation during GRACE gaps, and those earlier decisions alter the soil-water state inherited in later years. Their outcomes therefore combine rainfall-based selection with additional years of management. Matched coverage preserves the appropriate comparison for the incremental effect of storage information. Finer quota choices, alternative thresholds and verified release availability would define different decision experiments with their own selection and evaluation requirements.

## 4.5. Crop-response constraints and management reliability

Management selection depends on the modeled marginal response to water, in addition to absolute grain and canopy agreement. Intercepted-PAR radiation-use efficiency (Kiniry et al., 1989), cohort-specific leaf area and thermal senescence connect canopy development with carbon supply; grain number and fixed individual grain mass constrain the sink. Root uptake and the assimilation stress exponent then determine how water shortage changes production. Variation among wheat-model transpiration formulations demonstrates the sensitivity of water-use predictions to these processes (Cammarano et al., 2016). Water coefficients near their prescribed bounds remain effective empirical estimates whose physiological interpretation and uniqueness are limited.

The estimation and evaluation data provide different evidence for these responses. Earlier growth estimates use 480 wheat and 433 maize observations across four wheat sites and five maize sites, with whole site-years and balanced site contributions. Incomplete station irrigation histories limit transfer comparisons, and station observations have

no weight in the fitted water-response objective. Wuqiao 2016–2018 supplies the documented treatment seasons for that fit. Wuqiao 2019 supplies a single test year with reported total storage, assumed layer distributions and transferred soil hydraulics; prior inspection of its outcomes makes the test retrospective (Figures 3, 6 and 7). Regional comparison years also overlap crop-fitting years, so regional production differences characterize model scenarios rather than time-forward crop validation.

The absolute regional production reference also depends on the prescribed management and crop parameters. Conventional simulations produce 8.065 t $ha^{-1}$ of wheat dry grain and 8.691 t $ha^{-1}$ of maize dry grain per year. These values represent fixed irrigation dates, crop calendars and nutrition modifiers rather than an independently validated regional harvest inventory. The 0.732 t $ha^{-1}$ $yr^{-1}$ allocation advantage depends on the same simulated marginal responses; its regional scaling does not remove field-response uncertainty.

Substantial field-test errors constrain confidence in irrigation-response transfer. Wheat grain nRMSE was 83.27%, with a bias of +3.108 t $ha^{-1}$ at the published 13% moisture basis; the absolute W3−W0 ET-contrast error was 116.37 mm. Maize ET NSE was negative, with an ET-contrast error of 52.74 mm (Table 3; Table S5). These discrepancies affect the response functions used to select regional allocations. Numerical water-balance closure demonstrates conservation but supplies no measure of response accuracy. The eight purposive actual-cell checks address a separate uncertainty, spatial aggregation, and provide no regional field-validation sample. Depth-resolved soil water and complete dated management would better constrain uptake and redistribution; canopy and soil-moisture assimilation also requires treatment of retrieval errors and observation operators (Ines et al., 2013).

Production-retention targets introduce a further management distinction between period means and annual reliability. Mean retention of 96.27% and 99.63% coincided with annual minima of 92.12% and 98.58%; the 95% strategy met that threshold in eight of nine class-available years (Figure 9). A mean target consequently supplies no shortfall probability or guarantee for an individual farm. Local production floors, acceptable shortfall frequency and a reserve entering the next crop can complement regional delivery and consumption constraints. Forecast-based irrigation studies illustrate explicit treatment of water-stress risk (Roy et al., 2021). Field scheduling additionally requires current soil-water information, weather forecasts and application timing.

## 4.6. Limitations and future directions

The principal limitation is uncertainty in the crop responses used for strategy selection. Field testing covers a single, previously inspected year, while incomplete station management and assumed initial water profiles constrain transfer. Remaining errors in ET and grain responses prevent the simulated policy advantages from being

interpreted as independently validated field benefits. Further evaluation requires new site-years with complete management records, withheld from both model development and strategy selection. Concurrent measurements of soil water by depth, ET, canopy development and harvest would allow separate assessment of water redistribution, crop stress and irrigation-treatment contrasts.

Regional results also depend on a static rotation map, representative climate–soil units, prescribed irrigation dates and fixed nutrition modifiers. Checks at purposively selected source cells cannot bound regional uncertainty. Dynamic nitrogen cycling is absent, and changes in crop area or actual farm practices are not reconstructed. Future analyses should expand evaluation at individual source cells and propagate uncertainty in crop parameters, soil hydraulics, weather and spatial aggregation through paired policy histories. Allocation stability and the magnitude and persistence of local losses require uncertainty assessment alongside the regional mean.

The storage comparison is specific to the tested quota grid, classification thresholds and lower-storage conditions represented during later comparison years. GRACE release availability before sowing remains unverified, so the scenarios do not establish operational scheduling performance. Prospective decision experiments should use information available on each decision date, maintain matched rainfall-only controls and evaluate both storage states. Finer quotas and application timing informed by soil water could be tested with constraints on consumptive use, local production and the risk of production shortfalls.

Finally, the profile uses free drainage at its lower boundary, with no capillary supply or deeper subsurface storage, and field irrigation is not linked to measured abstraction. Groundwater applications require joint accounting of soil water and aquifer flows, constrained by observations of pumping, recharge and storage change. Delivery capacity, irrigation costs and farmer participation also require assessment before the agronomic scenarios can support operational management.

## 5. Conclusions

At an identical 50% irrigation reduction, targeted allocation retained 0.732 t $ha^{-1}$ $yr^{-1}$ more simulated combined wheat–maize dry grain than uniform cuts during 2014–2025. This reduced regional grain loss by 7.219 million t $yr^{-1}$ (4.87% of uniform production), while increasing crop-plus-fallow ET by 20.76 mm $yr^{-1}$ (2.048 $km^3$ $yr^{-1}$). Relative to uniform allocation, mean grain losses occurred on 37.60% of mapped rotation area and persisted in all twelve years on 21.34%. Regional production retention therefore requires separate consumption and local production constraints.

Annual strategies selected at 95% and 98% grain-retention targets retained 96.27% and 99.63% of conventional production over nine years with eligible GRACE observations. The 95% strategy reduced irrigation by 130.72 mm and ET by 36.84 mm per rotation. Matched rainfall-only controls selected identical quotas and produced identical outcomes, so storage classification provided no additional decision benefit within the tested quota design and monitoring coverage.

Continuous rotation accounting preserves the influence of preceding management on subsequent production and water use. The field-test errors and incomplete station management delimit confidence in transferring the estimated crop responses. Translation of these regional scenarios into irrigation management depends on observed response accuracy, explicit production-risk constraints and field scheduling informed by soil water and forecasts. Groundwater effects additionally require pumping, recharge and deeper subsurface storage beyond the crop-model boundary.

## Data and code availability

Model code, frozen calibration and regional inputs, observation subsets and quality exclusions, weather, calibrated parameters, predictions, evaluation metrics and figure-data tables are available at https://github.com/SmartAG-Team/agri_water_management. The repository includes a standalone Open Crop Model package, pinned dependencies, reproduction commands and SHA-256 provenance. Reproduction uses the included inputs without another model checkout or access to the historical workspace. Original datasets retain their provider licences and attribution requirements, documented in the repository source catalogue.

## Declaration of generative AI and AI-assisted technologies

AI assisted with grammar checking.

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